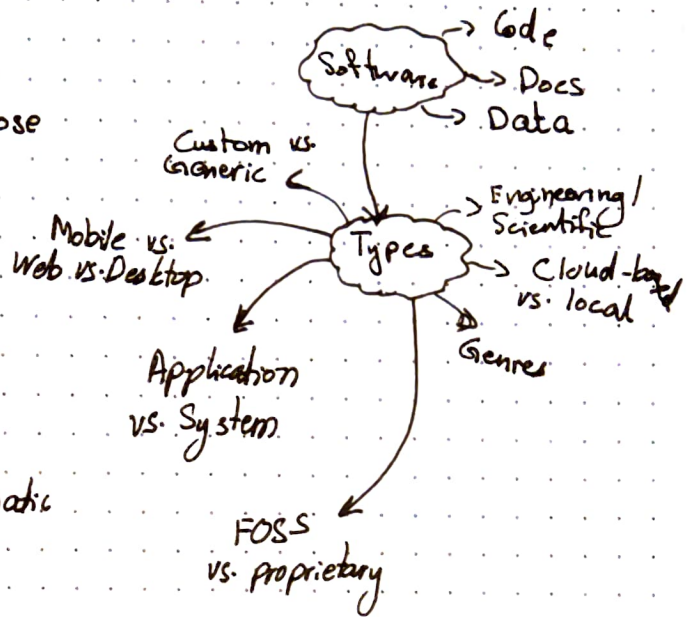
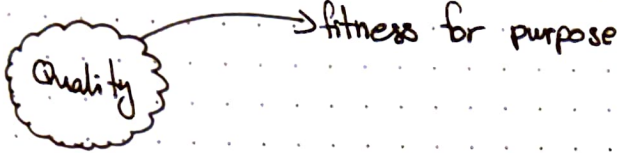


"Garbage in, garbage out."



Engineering: finding efficient solutions about for real-life problems in a systematic and efficient manner.

Problems compound. (snowball effect, domino effect).

ENIAC → Software → First CS dept. → Software Engineering
 1940s late 1940s # in US 1962

THRAL-25
 mid-late 1980s
 Denver, Colorado
 Airport catastrophe
 Washington prison malfunction: over 3000 prisoners released earlier.

Toyota recalled 600,000 'Prius' cars in 2016
 Nissan recalled ~3.5m Flex

WHY SQE is a separate field?

- nature (virtual/logical vs. physical)
- size
- team setting
- formal relation.
- interfaces
- personnel turnover

Algorithmic Morality

SQE LO3 28-08-2024

Software Quality Assurance Engineering

Quality

fitness for purpose

- conformance to explicit and implicit

technical and managerial requirements

Perception of Quality $\neq$ Quality

Functional
Non-functional
services offered
constraints

budget (cost)
schedule (time)

Software Quality Problems

- Error: 'to err is human' incorrect action or inaction.
- Fault: a state of the software. not all errors become faults.
Bug/Defect
- Failure: an event. not all faults become failures.
- Incident: detected ~~errors~~ failure. not all failures get noticed.

umbrella activity, process oriented

Software Quality Assurance: the set of systematic, planned activities required to provide adequate confidence in that the software produced through the software development process conforms to the explicit and implicit, technical and managerial requirements.

"A defect is a delivered fault."
- Pressman not for us

subset of SQA, product oriented

Software Quality Control

"An accident is a severe failure"

SQE LO4 30-08-2024

Capers Jones $\rightarrow$ Software Project Management

"Short-cutting a day of QA activity early in the project is likely to add 3 to 10 days of unnecessary activity downstream."

- Capers Jones

- The most important deliverable of Reqs. Engineering:



Quality Factors ^{dimensions, attributes, content groups, SRS headings}
-ilities

(Classic, 1977)
McCall's Quality Factor Model ^{An abstraction of the real thing.}
day-to-day operations ^{total 11 Quality Factors}
product operation (5) ^{change} product revision (3) ^{outside-world interaction} product transition (3)

- Correctness
- Maintainability
- Portability
- Reliability
- Flexibility
- Reusability
- Integrity
- Testability
- Interoperability
- Efficiency
- Usability

Correctness: functional requirements, output-mission, output accuracy, output completeness, output up-to-date-ness, availability (response time), standards

Reliability: failure rates, probabilities

critical systems

Integrity: security (Authentication & Authorization)

Efficiency: optimal use of hardware resources (CPU, RAM, storage, communication bandwidth, battery life)

Usability: ease-of-use

- operations
- training

Question 03 (b)

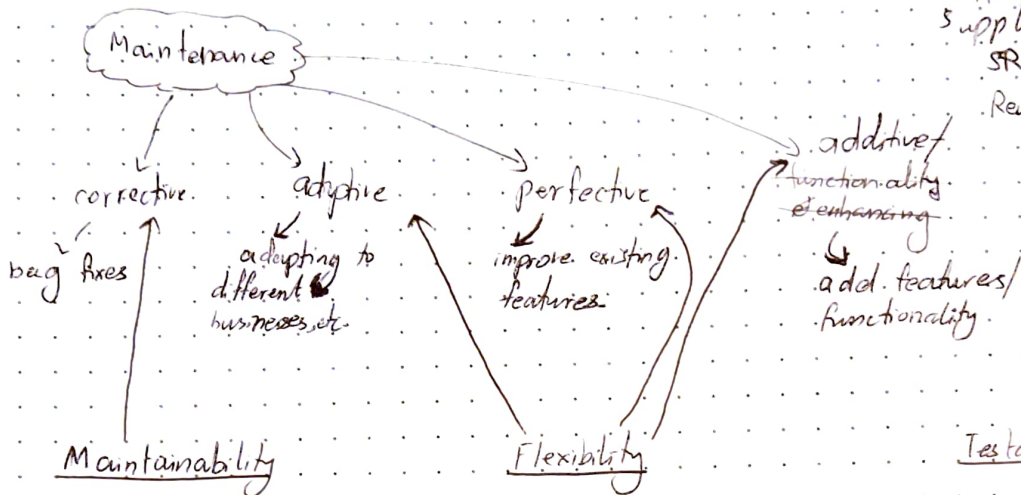
SQE LOS 04-09-2024

BIOS does POST

McCall's QF Model

Power-On Self Test

Product Revisers change and maintain



Testability

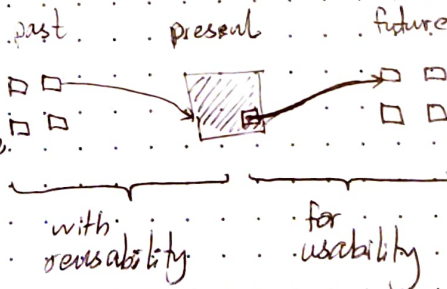
- easy to test
- intermediate results or logs
- allow operators and maintainers to test and diagnose the software.
- both comments and readability are important
- comments are not an excuse for bad programming practices.

Product Transition:

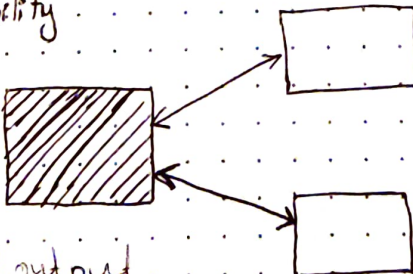
- Portability: cross-platform support.

- Reusability
- for
- with

- improve documentation
- increase cohesion
- decrease coupling



- Interoperability



- standard output for mats

Alternative QF Models.

- Evans & Marciniak (1987)
- Deutsch & Willis (1988)
- Verifiability $\rightarrow$ mathematical verifications
- Expandability $\approx$ Flexibility
- Safety
- Survivability $\approx$ Reliability
- Manageability

human safety
 $\rightarrow$ removing unsafe condition

$\rightarrow$ administrative tools for change management.

Project Managers, UI/UX, programmers, testers, maintainers, dev ops, etc.
 ↳ many people can make mistakes.

Components of SQA system

Pre-Project Components

→ Contract Review.

RFP → Request for Proposal.

Draft → Draft Review

SC evaluates submissions

↳ one gets a contract.

↳ contract draft → review

→ development & quality plans.

→ cost, resource, estimations

→ WBS, Risk plans

Life-cycle components

→ Reviews

- team-based meetings

- variable degree of formality

- types: formal technical review (most formal),

↳ FTR, Design Reviews, FDR

↳ done by senior people

→ Testing

- regression, smoke, automated, manual, unit, integration, other tests.
 black box, white box.

→ Expert. Opinions

- outside reviews, etc.

- Third Party

SS₁

SS₂

SS - SC

COTS

ROTS, GOTS

Commercial
off-the-shelf
components.

SQA applies to maintenance as well

Component Categories

Applicable
to projects at
hand

Re-project

Life-cycle

Infrastructure

Management

Standards

Human

Applicable
organization-
wide

SS

SC

↳ contract

Problem
Analysis
Phase

Solution
Phase

Inc. RE A D J J D/O O-M DR

Part 1

Part 2

Inspections, Walk Through &
(least formal)

Peer Reviews

SCPE L06 (Continued)

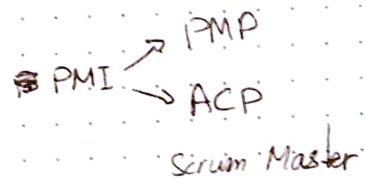
Infrastructure Components

- Procedures & Work Instructions
- Supporting Quality Devices
 - Templates (SRS, project plan, risk plan, etc.)

- Training & Certifications
 - Training improves

- Software Configuration Management
 - Version control
 - Git, subversion, etc.

- Analysis of Post-Projects
 - what went wrong
 - where most mistakes occur



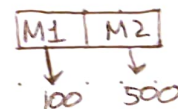
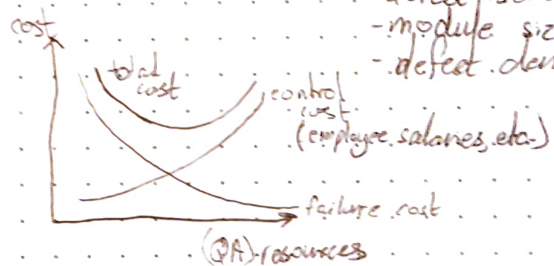
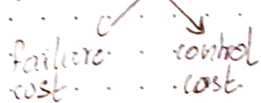
Management Components

- Progress Control
 - Managers do planning, monitoring, controlling

- Quality Metrics
 - defect density
 - productivity
 - defect severity → weighted defects
 - module size

- Cost

- Minimize project cost



Standards Components

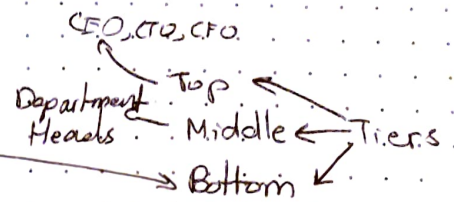
- Quality Management (What)
 - CMMI capability Maturity Model Integrated
 - developed by CMU

- ISO 90003

- Project Process (How)

- IEEE 1012
- ISO/IEEE/IEC 12207

Human Components
→ Employees, team members, project managers.



Factors Affecting QA system

- organization size $\propto$ QA effort required.

- project complexity $\propto$ QA effort required.

- experience (relevant to the project domain) $\propto$ $\frac{1}{\text{QA effort required}}$

- 3rd party engagement $\propto$ QA effort required.

- acquaintance of team members with each other $\propto$ $\frac{1}{\text{QA effort required}}$

- extent of reuse (with & reuse) $\propto$ $\frac{1}{\text{QA effort required}}$

- qualification $\propto$ $\frac{1}{\text{QA effort required}}$

SQE LOT 108 11-09-2024
13-09-2024

BRING CALCULATOR usp (Unique Selling Point)

INTEGRATION OF SQA ACTIVITIES IN SDLC

Software Processes

Process: a set of activities approached in a specific order.
→ approach, roadmap,
Process Model: an abstraction of the process.

Spiral is a meta-process

Usability of drop-down
vs. text box
Recognition vs. Recall

Process Control vs.
Quality Management:
What vs. How,
not product vs.
process oriented

change-friendliness

Conventional,
Rigorous, Heavy-weight

I Know It When
I See It
Syndrome

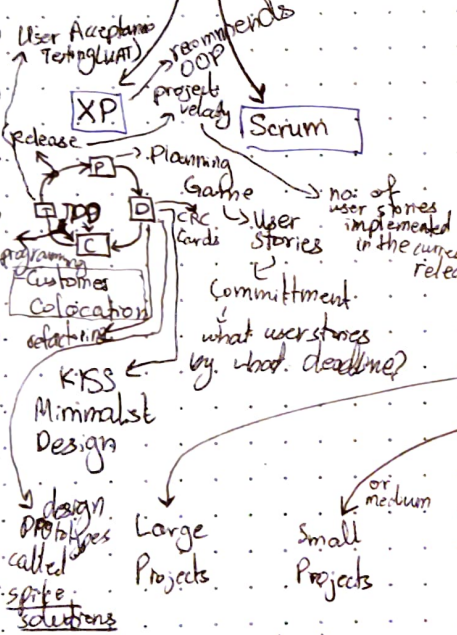
Processes

Agile

Traditional

Evolutionary

Non-Evolutionary



Throw-away prototypes

Build-upon prototypes

no. of user stories implemented in the current release

Planning Problem: when will it end

only in hybrid versions

Change welcome

High Expertise

High Manpower

Naturally modularizable Problem

Large Projects

Small Projects

Stable Reqs.

Unstable Reqs.

Prototyping

IKIWISI

Prototyping

Heavy weight

risk-driven process

Waterfall (classical)

large projects, stable reqs., inacceptable when: - unscope, customer volatile, reqs. feedback needed

only in hybrid versions

Change welcome

High Expertise

High Manpower

Naturally modularizable Problem

Large Projects

Small Projects

Stable Reqs.

Unstable Reqs.

Large Projects

Incremental (Pure)

no. of Features

Incremental (Pure)

Incremental (Pure)

Incremental (Pure)

Incremental (Pure)

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Incremental (Pure)

Incremental (Pure)

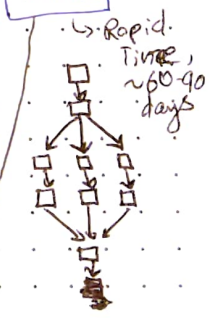
Incremental (Pure)

Incremental (Pure)

Incremental (Pure)

Incremental (Pure)

Incremental (Pure)



Rapid Delivery Required

- Kent Beck's book on XP
- Refactoring by Martin Fowler

User Stories

As a who, I want what so that why

- Priorities (Sc) ↓ value
- Cost (\$)

February 2001, 17 people signed the agile manifesto in Utah.

Kent Beck, Robert C. Martin

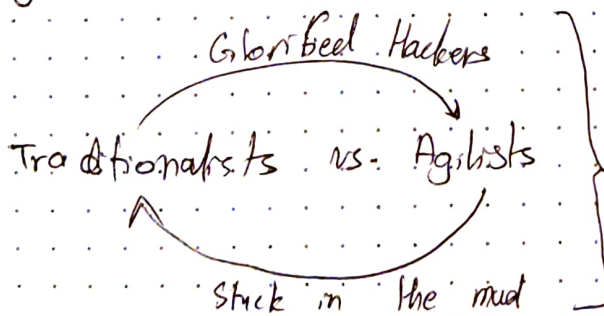
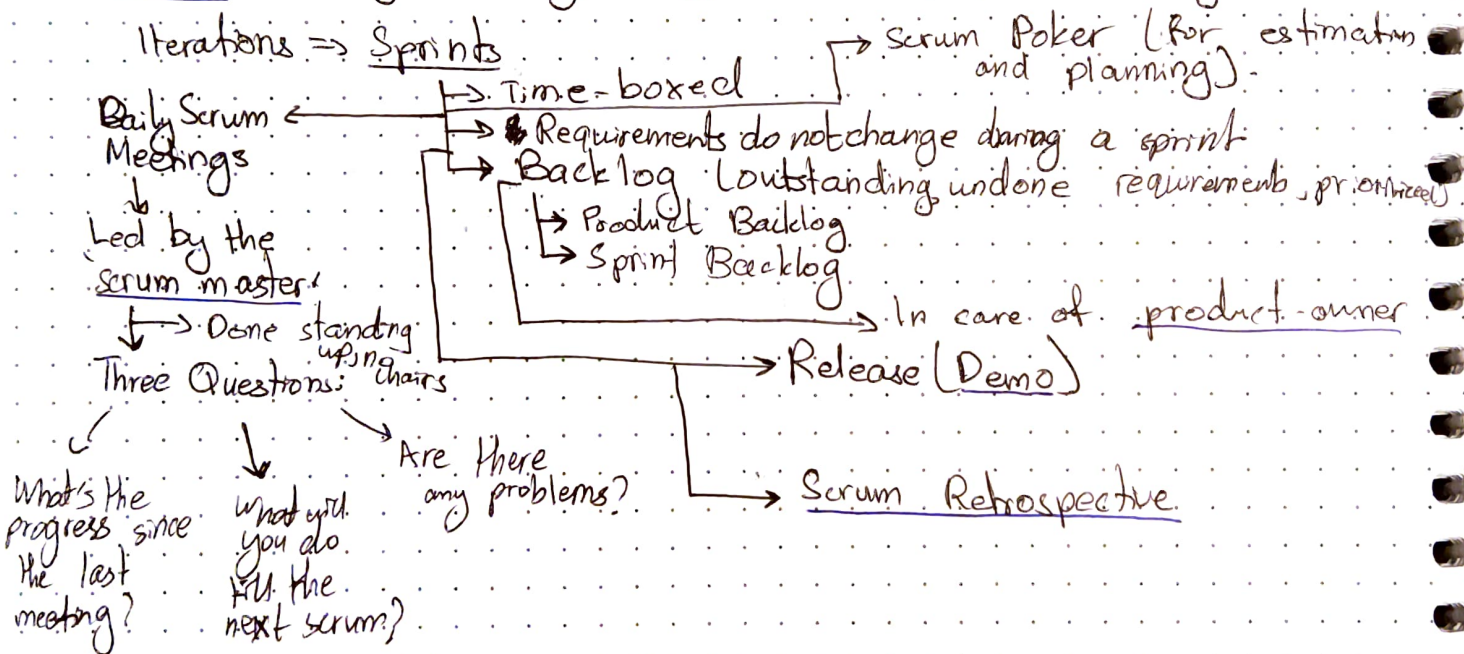
→ later extended to other fields like manufacturing, design, and project management.

Agile Manifesto

- | | | | |
|----|-----------------------------|-----|-----------------------------|
| 1- | Individuals & Interactions. | vs. | Processes & tools |
| 2- | Working software | vs. | Comprehensive Documentation |
| 3- | Customer Communication | vs. | Contract Negotiation |
| 4- | Responding to change | vs. | Following a plan |

• Scrum managerial-heavy, as opposed to technical-heavy as in XP.

Iterations ⇒ Sprints



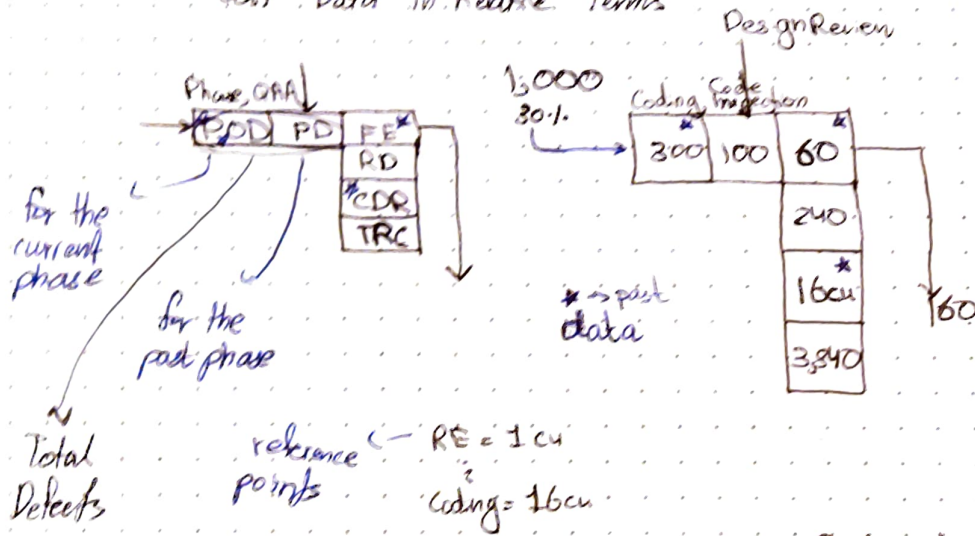
Barry Boehm → Balancing Agility and Discipline

Dimensions of SQA

Defect Removal Effectiveness & Cost Model

- QA activities are filters, not 100% accurate, but still cost something.

- Defect
- Filtering Effectiveness
- Cost Data in Relative Terms



$$RD = (LPDD + PD) \times \frac{FE}{100}$$

$$TRC = RD \times CDR$$

Assumptions:

- All SQA activities act like filters
- Past data is accurate and still relevant.

Verification: are we making the product right?

Validation: are we making the right product?

Qualification:

- coding conventions
- good SE practices
- general standards

Circle Area Examples

- Verification: is it programmed as TR?
- Validation: Did the customer actually need this?

- Phase Originated Defects (POD)
- Past Defects (PD)
- Filtering Effectiveness (FE)
- Removed Defects (RD)
- Cost of Defect Removed (CDR)

Total Remaining Cost (TRC)

Question 03(b)

SQE (100)

No.	Software Development Phase	originating defects in this phase
1	Requirements	15%
2	Design	35%
3	Coding (30% coding, 10% integration)	40%
4	Documentation	10%

Standard QA plan (Plan A) → Example: 1773.3 cu total

No.	QA activity	Filtering effectiveness	Cost of removing defects
1	Requirement specification review	50%	1 cu
2	Design review	50%	2.5 cu
3	Unit Test - Code	50%	6.5 cu
4	Integration test	50%	16 cu
5	Documentation Review	50%	16 cu
6	System Test	50%	40 cu
7	Operations phase	100%	110 cu

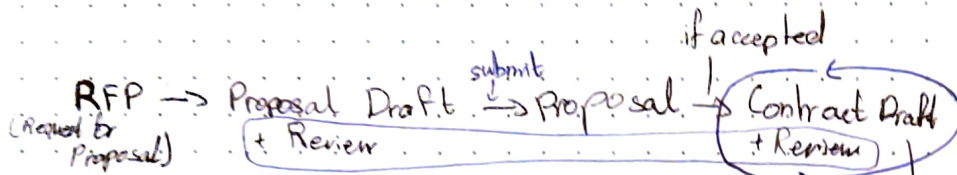
Comprehensive SQA Plan → Example: 1046.4 cu total

No.	QA Activity	Filtering Effectiveness	Cost of Removing Defects
1	Requirement specification review	60%	1
2	Design inspection	70%	2.5
3	Design review	60%	2.5
4	Code inspection	70%	6.5
5	Code - review unit test	40%	6.5
6	Integration test	60%	16
7	Documentation review	60%	16
8	System test	60%	40
9	Operations Phase	100%	110

Contract Review

Pre-project component

Crime has consequences



- Requirements clarified
 - ↳ scope
- Alternative Approaches
- Relationship Formality
 - Focal Person representative
 - Project deliverables
 - Change approval process
- Risks
 - technological
 - non-technical (personnel turnover, economic, political conditions)
- Estimates
 - ↳ estimation techniques
 - Effort
 - Duration
- Capacity / Capability
 - Can the supplier do it?
 - Can the consumer pay us?
- Participants
- Intellectual Property
 - Who gets to own the copyrights.

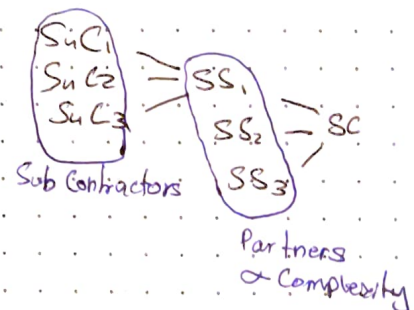
a legal document
sign
Contract
Formality benefits both sides by enforcing protocols.

SS — SC

Legal contract
- handshake
- both parties are signatories

SS ~ SC blurred when
- SS and SC belong to the same org.
- embedded systems
- products for customers

↳ SCA activities should be scheduled.



Goals of Contract Draft Reviews

- no unclarified issues
- understandings documented
- no "new" changes

no ambiguity
no less
no more

extent of the contract draft review

- depends on project size, complexity, organizational complexity, acquaintance/familiarity/expertise

Characteristics of Bad Contracts

- unrealistic estimates
- loose requirements

estimates can vary based on the software development process. But the choice of the process need not be documented.